Md. Bipul Islam

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GitHub | LinkedIn | Google Scholar | ORCID | ResearchGate | Portfolio

Research Interests

Trustworthy AI for Healthcare, ML, DL, Computer Vision, NLP, LLMs, XAI, Cybersecurity

Academic Credentials

M.Sc. (Engineering) in Computer Science and Engineering Dec 2026 (Expected)
Hajee Mohammad Danesh Science and Technology University, Dinajpur, Bangladesh
Research Title: "Self-Supervised Contrastive and Masked Representation Learning for Reliable Arrhythmia Classification Using Noisy Multi-Lead ECG Signals"
Supervisor: Dr. Ashis Kumar Mandal
CGPA: **3.67 / 4.00** (Up to Final Semester)

B.Sc. (Engineering) in Electrical and Electronic Engineering December 2019
Hajee Mohammad Danesh Science and Technology University, Dinajpur, Bangladesh
CGPA: **2.92 / 4.00**

Research Experience

Student Researcher, Elite Research Lab *Jul 2026 – Present*

- Conducting research on AI in Healthcare, Clinical AI, and Human-Centered AI, focusing on developing reliable and interpretable models for clinical decision support.

Graduate Research Assistant, Computer Science Lab, HSTU *Jan 2025 – Present*

- Developing self-supervised contrastive and masked representation learning methods for reliable arrhythmia classification from noisy multi-lead ECG signals using PyTorch and TensorFlow.
- Leading a Bangladesh Government funded research project on AI-based automated detection of potato leaf diseases with actionable farmer guidance.

Research Assistant, Walton ICT Lab (R&D Lab), Walton *Mar 2023 – Dec 2024*

- Researched and developed cross-platform mobile and IoT software products, including educational systems, HR management tools, and smart device applications.

Manuscripts in Preparation

[1] **Md. Bipul Islam** and Ashis Kumar Mandal, "Beyond Accuracy: Calibration and Explanation Reliability of Contrastive, Masked, and Hybrid Self-Supervised ECG Models for Arrhythmia Classification under Noise, Lead Loss, and Dataset Shift." *M.Sc. Research – in progress.*

[2] **Md. Bipul Islam**, Asiqul Tareq Titu, and Ashis Kumar Mandal, "AI-Based System for Automated Detection of Potato Leaf Diseases and Actionable Farmer Guidance." *Bangladesh Government Funded Research – manuscript in preparation.*

[3] **Md. Bipul Islam**, Jyotirmay Debnath Pulak, and Ashis Kumar Mandal, "Explainable Physics-Informed Neural Networks for Robust Health Risk Prediction Using Multi-Condition Medical Tabular Data." – *manuscript in preparation.*

[4] Dino Nath Roy, **Md. Bipul Islam**, and Ashis Kumar Mandal, "Interpretable Brain Tumor Segmentation Using Foundation-Model Adaptation with Voxel-Level Uncertainty Quantification on Multi-Modal MRI." – *manuscript in preparation.*

[5] **Md. Bipul Islam**, Rafsan Mahmud, Shirin Mahbuba, and Tazin Alam, "Early Warning of Model Degradation Under Progressive Image Corruption: A Comparative Analysis of Confidence, Behavioral, and Explanation Instability." – *manuscript in preparation.*

Teaching Experience

Teaching Assistant, Object Oriented Programming (CSE 201) *Jan 2025 – Present*
Department of Computer Science and Engineering, HSTU

- Assist Dr. Ashis Kumar Mandal in delivering lectures and lab sessions on Object Oriented Programming to undergraduate students.
- Prepare course materials, evaluate assignments, and conduct tutorial sessions covering classes, inheritance, polymorphism, and encapsulation.

Private Tutor *Jan 2016 – Dec 2024*

- Taught programming and professional Android/iOS app development to undergraduate students, guiding several toward successful employment after their graduation.
- Mentored HSC-level students in Higher Mathematics, ICT, and English — building conceptual foundations and guiding board-exam and university-admission preparation.
- Mentored SSC-level students in Physics, Chemistry, Mathematics, ICT, and English — strengthening fundamentals through concept-based teaching, regular assessments, and personalized guidance.

Work Experience

Software Engineer (Android & iOS) – Team Leader *Mar 2022 – Dec 2024*
Walton Digi-Tech Industries Ltd, Dhaka, Bangladesh

- Led development of cross-platform apps, including Bench Educational App and HRMS, for internal use and clients.
- Built IoT-based Android apps for Walton Tick Smart Watch and FitPro Smart Weight Scale, enabling real-time device communication and data analytics.
- Managed app lifecycle, REST API integration, cloud sync, and mentored junior developers.

Android Application Developer *Jan 2020 – Feb 2022*
Hovata Technologies, Dhaka, Bangladesh

- Developed IoT-based Android apps including Digital Petrol Pump (Petrol ERP and POS), Smart Parking Management System (Hovata Parking), and Smart Wallet Management App (Linn Books).
- Implemented real-time device communication, secure transactions, and cloud integration for digital solutions.

Technical Strengths

- **AI & Data Science:** NumPy, Pandas, Matplotlib, Seaborn, SciPy, Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, spaCy, OpenCV, PySpark, Keras, NLP, DASK, XAI
- **Research & Data Tools:** Jupyter Notebook, Google Colab, LaTeX, MATLAB, R, Excel, SPSS, Minitab, TORA, NetworkX, ggplot2, Plotly, Zotero, Mendeley
- **Software Development:** Android App (Java, Kotlin, Flutter), iOS App (Swift, Flutter)
- **Backend Development:** Node.js, Spring Boot (Java, Kotlin), Ktor (Kotlin), Django, FastAPI, Flask

Standardized Test Scores

GRE General Test (February 2024): Quantitative – 159, Verbal – 146, AWA – 3.5, Total – **305/340**

Duolingo English Test (January 2024): Overall – **120/160** (Literacy – 115, Comprehension – 135, Conversation – 110, Production – 95)

Projects

Complete list available at: bipulhstu.github.io/projects

Sentiment Analysis on Social Media Data

- Analyzed social media posts to classify sentiment (positive, negative, neutral) using NLP preprocessing (spaCy, NLTK) and transformer models (BERT, RoBERTa).
- Achieved high F1-scores with strong generalization across diverse topics.

- **Tools:** Python, Hugging Face Transformers, spaCy, NLTK
- **Links:** GitHub | Live Demo

Anomaly Detection in Network Traffic

- Detected anomalies in network traffic for cybersecurity, performance monitoring, and compliance using Isolation Forest (unsupervised ML) with temporal feature engineering.
- Achieved accurate detection of unusual traffic patterns with low false positives; deployed via Streamlit dashboard.
- **Tools:** Python, Scikit-learn, Pandas, Streamlit, Plotly
- **Links:** GitHub | Live Demo

Bangladeshi Flower Image Classification using Deep Learning

- Built a deep learning web app for classifying 13 types of Bangladeshi flowers using transfer learning with MobileNetV2 (fine-tuned last 50 layers) and data augmentation.
- Achieved 99.87% validation accuracy with real-time classification and confidence scores.
- **Tools:** Python, TensorFlow/Keras, Streamlit, OpenCV, PIL, NumPy, Pandas
- **Links:** GitHub | Live Demo

Relevant Certifications

All certificates available at: bipulhstu.github.io/certifications

- AI for Medicine Specialization (Coursera – DeepLearning.AI)
- Mathematics for Machine Learning and Data Science Specialization (Coursera – DeepLearning.AI)
- Deep Learning Specialization (Coursera – DeepLearning.AI)
- Machine Learning Specialization (Coursera – Stanford University / DeepLearning.AI)
- IBM Machine Learning Professional Certificate (Coursera)
- Python 101 for Data Science (IBM)
- Generative AI for Everyone (Coursera – DeepLearning.AI)
- AI for Everyone (Coursera – DeepLearning.AI)
- Machine Learning Scientist in Python (DataCamp)
- Data Scientist Certification (DataCamp)

Leadership and Voluntary Activities

- **Public Relations Executive,** IEEE CS BDC GSMF — June 2026 – Present
- **Class Representative,** M.Sc. in Computer Science — January 2025 – Present
- **Mobile App Development Team Leader** — April 2023 – Dec 2024
- **Member,** IEEE HSTU Student Branch — 2017
- **Executive Member,** Skill Development Club of HSTU — 2017–2018
- **Member,** EEE Club of HSTU — 2015–Present
- **General Secretary,** Prothom-Alo Bondhushava, HSTU — 2019
- **IT Secretary,** HSTU-OSN — 2016–2017
- **Program Manager,** Programmers Arena — 2016–2018

References

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